A THEORY OF INTELLIGENCE AND PRODUCTION

From Ideas to Action

Napoleon, Berthier, and Lannes



Research model | October 2026

MOTIVATION

Where will AI's economic value come from?

VIEW 1 · BETTER IDEAS

AI generates and selects ideas that humans would not have conceived.

Value comes from shifting the frontier of what is worth trying.

VIEW 2 · BETTER REALIZATION

AI makes the existing stock of human ideas faster and cheaper to turn into workable plans.

Value comes from closing the gap between an idea and its best plan.

This deck builds a production model that separates the two views, and adds the stage neither view controls: **physical execution**, which may bound throughput however good the thinking upstream.

THE HYPOTHESIS

Intelligence is not a single factor of production

NAPOLEON

Conception
What is worth trying?

Produces and evaluates a vague idea ξ.

BERTHIER

Mental execution
What plan implements it?

Formulates the problem and finds a *fallible* plan $\hat{a}$.

LANNES

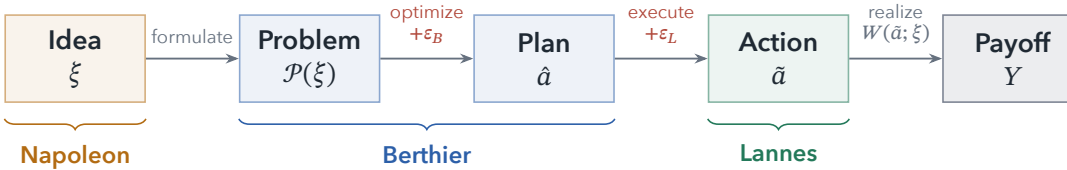
Physical execution
What actually happens?

Carries out the plan, possibly imperfectly: $\tilde{a}$.

Working hypothesis. Current AI chiefly improves **Berthier**; an ASI would also be an exceptional **Napoleon**. **Lannes** remains necessary whenever output requires physical action.

THE PROJECT PIPELINE

An idea is not a plan; a plan is not an outcome



> An idea ξ specifies a desired possibility. **Berthier** converts it into a well-defined problem, finds a plan, and hands the plan to **Lannes**. The resulting physical action generates payoff Y. Errors enter at planning (ε_B) and at execution (ε_L).

Napoleon creates the variation across projects

IDEA-GENERATING TECHNOLOGY

$$\xi_i \overset{\text{i.i.d.}}{\sim} F_A, \qquad \lambda_A = \frac{n_A}{\tau_A}$$

- One idea per Napoleon every τ_A.
- n_A Napoleons work in parallel.
- F_A governs the *quality* of ideas, not just their number.

VALUE OF AN IDEA

The value achievable with an optimal plan and perfect execution:

$$v(\xi) = \max_{a \in \mathcal{A}(\xi)} W(a; \xi)$$

In the baseline, Napoleon observes $v(\xi)$ and decides whether the idea deserves downstream capacity.

Progress at this stage can raise the arrival rate λ_A, shift the distribution of $v(\xi)$ upward, or sharpen Napoleon's ability to recognize valuable ideas.

Berthier turns a vague idea into a fallible plan

THE BENCHMARK PROBLEM INDUCED BY AN IDEA

$$\mathcal{P}(\xi): \quad \max_{a \in \mathcal{A}(\xi)} W(a;\xi), \qquad a^*(\xi) \in \arg\max_{a \in \mathcal{A}(\xi)} W(a;\xi)$$

$$\boxed{\hat{a} = a^*(\xi) + \varepsilon_B} \qquad \mathbb{E}[\varepsilon_B \mid \xi] = 0, \quad \mathrm{Var}(\varepsilon_B \mid \xi) = \Omega_B$$

FORMULATION

Translate a vague objective into decision variables, constraints, and payoffs.

OPTIMIZATION

Spend scarce planning time τ_B searching for a good solution.

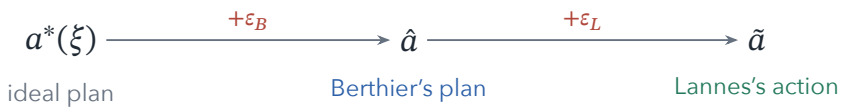
WHAT AI CHANGES

A better Berthier lowers Ω_B, τ_B, or both.

ε_B pools formulation and solution errors; a systematic formulation mistake would add bias (Appendix A2).

A correct plan does not guarantee correct execution

$$\boxed{\tilde{a} = \hat{a} + \varepsilon_L} \qquad \mathbb{E}[\varepsilon_L \mid \xi, \varepsilon_B] = 0, \quad \mathrm{Var}(\varepsilon_L \mid \xi) = \Omega_L$$



Sources of ε_L: late arrival, miscommunication, local friction, inaccurate maneuvering, or inability to supply the commanded effort.

A single realization of ε_L may happen to offset Berthier's mistake; the model guarantees losses only in expectation.

Planning and execution losses enter additively

QUADRATIC LOSS AROUND THE IDEAL PLAN

$$W(a;\xi) = v(\xi) - \tfrac{1}{2}\big(a - a^*(\xi)\big)^{\top} H\big(a - a^*(\xi)\big), \qquad H \succ 0, \qquad Y(\xi) = W(\tilde{a};\xi)$$

The realized deviation is $\varepsilon_B + \varepsilon_L$. With mean-zero, independent errors, the cross term vanishes:

$$\mathbb{E}[Y \mid \xi] \;=\; v(\xi) \;-\; \underbrace{\tfrac{1}{2}\operatorname{tr}(H\Omega_B)}_{\kappa_B\text{: planning loss}} \;-\; \underbrace{\tfrac{1}{2}\operatorname{tr}(H\Omega_L)}_{\kappa_L\text{: execution loss}}$$

Baseline: H, Ω_B, and Ω_L do not depend on ξ, so ideas can be ranked by $v(\xi)$ alone.

The slower downstream stage caps throughput

Stage	Workers	Time per project	Rate
Napoleon: ideas	n_A	τ_A	$\lambda_A = n_A/\tau_A$
Berthier: plans	n_B	τ_B	$\mu_B = n_B/\tau_B$
Lannes: actions	n_L	τ_L	$\mu_L = n_L/\tau_L$

With a sufficient supply of accepted ideas, long-run throughput is bounded by downstream capacity:

$$K = \min\{\mu_B,\ \mu_L\}$$

The latency of a single project is $\tau_A + \tau_B + \tau_L$; throughput is higher because different projects occupy different stages at the same time.

Scarce capacity justifies rejecting good ideas

Let $V = v(\xi) \sim G_A$, observed by Napoleon. Accept an idea iff $V \geq z$; each accepted idea uses one slot at both later stages. With $\kappa = \kappa_B + \kappa_L$, the planner solves

$$\max_z \; y(z) = \lambda_A \int_z^{\infty} (v - \kappa)\, dG_A(v) \qquad \text{s.t.} \qquad Q(z) \equiv \lambda_A \left[1 - G_A(z)\right] \;\leq\; K$$

OPTIMAL CUTOFF

$$z^* = \max\Big\{\underbrace{\underline{\kappa}}_{\text{break-even}},\; \underbrace{G_A^{-1}\Big(1 - \min\Big\{1, \frac{K}{\lambda_A}\Big\}\Big)}_{\text{capacity rationing}}\Big\}$$

The first term discards projects that lose value in expectation; the second reserves scarce capacity for the best ideas, so Napoleon rejects some *positive-value* ideas.

Assumes a continuous value distribution and constant per-project stage times.

Faster plans and better plans pay off differently

SPEED

$$\tau_B \downarrow \Longrightarrow \mu_B \uparrow$$

More ideas proceed to execution, *but only if Berthier is the binding stage*.

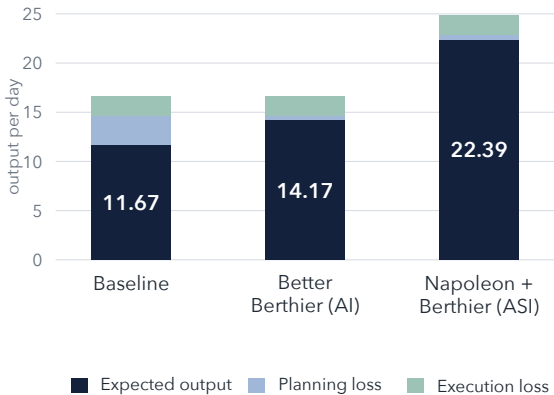
ACCURACY

$$\Omega_B \downarrow \Longrightarrow \kappa_B \downarrow$$

Each executed project yields more, *even when Lannes is binding*.

Envelope result. $\partial \mathcal{Y}^*/\partial \kappa_B = -Q^*$. A one-unit fall in Berthier's expected loss raises value by the flow of accepted projects.

Lannes binds, yet better thinking still adds value



	Ideas λ_A	Plans μ_B	Actions μ_L
Baseline	6	3	2
AI	6	8	2
ASI	9	8	2

Lannes completes only **two projects per day** in every scenario. Gains come from executing *more valuable* projects with *fewer planning errors*.

Baseline: $V \sim U[0, 10]$, $\kappa_B = 1.5$, $\kappa_L = 1$. AI: $\kappa_B = 0.25$. ASI: also $V \sim U[0, 14]$, $\lambda_A = 9$. The top K/λ_A share of draws is accepted. Illustrative parameters, not a forecast.

AI and ASI act on different margins

AI · THE BERTHIER MARGIN

- Faster planning: $\tau_B \downarrow$
- Better planning: $\Omega_B \downarrow$
- Same distribution of ideas F_A

ASI · ALSO THE NAPOLEON MARGIN

- More ideas: $\tau_A \downarrow$
- Better ideas: F_A shifts up in value
- Retains strong planning: τ_B, $\Omega_B \downarrow$

Main tension. As long as Lannes is required, cognitive intelligence faces a physical throughput bound. Idea quality and planning accuracy still matter *within* that bound.

Extension: time can buy accuracy at each stage

Suppose that, for one project, more time lowers each stage's expected loss:

$$\kappa_B(t_B) = \frac{b}{t_B}, \qquad \kappa_L(t_L) = \frac{\ell}{t_L}, \qquad b, \ell > 0.$$

With a fixed time budget $T = t_B + t_L$, choose the split:

$$\min_{t_B, t_L > 0} \frac{b}{t_B} + \frac{\ell}{t_L} \quad \text{s.t.} \quad t_B + t_L = T$$

OPTIMAL SPLIT

$$\frac{t_B^*}{t_L^*} = \sqrt{\frac{b}{\ell}}$$

An AI improvement that lowers b cuts the planning time needed for a given accuracy.

In the full pipeline the optimal allocation also depends on relative stage capacity and on the value of accepted projects.

MEASUREMENT

Lannes supplies a benchmark, not identification

WHAT CAN BE OBSERVED

- Berthier's submitted plan $\hat{a}$
- Lannes's realized action $\tilde{a}$
- The realized payoff Y

WHAT CAN BE LEARNED

$\tilde{a} - \hat{a}$ measures execution deviation.

$\hat{a} - a^*(\xi)$ measures planning deviation, *if a credible benchmark optimum exists*.

Repeated outcomes help test whether a Napoleon's ideas were good.

Physical outcomes alone do not separate poor ideas, poor plans, and execution noise. Benchmarks or additional variation are needed.

SUMMARY

Takeaways

$$y^* = \underbrace{Q^*}_{\text{accepted flow}} \times \Big(\underbrace{\mathbb{E}[V \mid V \geq z^*]}_{\text{value of selected ideas}} - \underbrace{\kappa_B}_{\text{planning loss}} - \underbrace{\kappa_L}_{\text{execution loss}} \Big)$$

1. **Intelligence has distinct margins.** Conceiving, planning, and executing each have their own capacity and error.
2. **The binding stage decides what pays.** When Lannes binds, faster planning adds nothing; only more accurate planning and better selection raise output.
3. **Accuracy is worth the flow of projects.** Each unit of planning-loss reduction raises value by Q^*.
4. **Better ideas work through selection.** More and better draws raise the quality of the K projects that capacity allows.

Open questions for the theory

1. **Endogenous complexity.** Do Napoleon's most ambitious ideas need more Berthier time, more Lannes time, or carry larger error variances: $\tau_B(\xi)$, $\tau_L(\xi)$, $\Omega_B(\xi)$, $\Omega_L(\xi)$?
2. **Occupational choice and wages.** How does cheaper Berthier labor change demand for Napoleon-like and Lannes-like workers?
3. **Learning and evaluation.** Can feedback from executions improve idea selection and distinguish failures of conception from failures of implementation?

> **Central question.** Does AI's first-order economic effect come from generating better ideas, or from making the existing stock of human ideas much easier to realize?

APPENDIX

A1. Closed form with uniform idea values

Assume $V \sim U[0, \bar{v}]$, constant expected losses $\kappa = \kappa_B + \kappa_L$, and downstream capacity K.

$$z^* = \max\left\{\kappa,\ \bar{v}\left(1 - \min\left\{1, \frac{K}{\lambda_A}\right\}\right)\right\}$$

The accepted flow and its average potential value are

$$Q^* = \lambda_A\left(1 - \frac{z^*}{\bar{v}}\right), \qquad \mathbb{E}[V \mid V \geq z^*] = \frac{\bar{v} + z^*}{2},$$

so the optimized output rate is

$$\boxed{y^* = Q^*\left(\frac{\bar{v} + z^*}{2} - \kappa\right)}$$

APPENDIX

A2. Beyond independent, unbiased errors

Let $\mathbb{E}[\varepsilon_B] = b_B$, $\mathbb{E}[\varepsilon_L] = b_L$, and $C_{BL} = \mathrm{Cov}(\varepsilon_B, \varepsilon_L)$. Then

$$\mathbb{E}[Y \mid \xi] = v(\xi) - \frac{1}{2}(b_B + b_L)^{\mathsf{T}} H (b_B + b_L) - \frac{1}{2}\,\mathrm{tr}\big[H(\Omega_B + \Omega_L + C_{BL} + C_{BL}^{\mathsf{T}})\big]$$

- **Biased planning.** Berthier may systematically misunderstand Napoleon's idea.
- **Correlated error.** Lannes may correct, amplify, or respond to Berthier's mistakes.
- **Uncertain outcomes.** Additional payoff shocks leave expected output unchanged if mean-zero and independent.